ENGR 470 Capstone Design Project I
Fall 2025
ENGR 470 | Tue/Thur 6:30 pm-7:45 pm | HWWE 208 & Virtual
Zoom Meeting ID: 554 616 3235 https://cofc.zoom.us/j/5546163235

Instructor: Qian Zhang, PhD
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Office Hours: Thur 11:30-12:30 pm

COURSE DESCRIPTION
This course is the first part of the Capstone Design Project. Students collaborate with CofC faculty and/or industry clients to develop a design project. The course integrates engineering knowledge, problem-solving skills, project planning, oral presentations, report-writing, ethical considerations, and teamwork skills. Student teams will present their project ideas before the Senior Design Faculty Committee, receive feedback, and refine their proposals into a final draft. All team projects are subject to instructor's approval.

Prerequisite(s): Senior Standing; Engineering Major

COURSE-SPECIFIC LEARNING OUTCOMES
Primary objectives of the course:
Students will develop an engineering project through research and application of methodologies learned in prior coursework. They will create written reports, and oral presentations, and will be evaluated through quiz, exam, project deliveries and participation in class activities.

STUDENT LEARNING OUTCOMES

- Identify an appropriate project, design a proposal, and develop a project plan addressing the needs and implementation of the project.
- Incorporate appropriate technical, ethical, safety, environmental, and sustainability standards into the capstone project.
- Validate and test project solutions within realistic constraints.
- Improve communication skills through technical writing and presentations.
- Utilize academic and industry references for project design.
- Gain experience in industrial project management, including planning, execution, and monitoring.
- Develop entrepreneurial skills, fostering innovation and commercialization of projects.
- Enhance professional readiness with resume preparation, mock interviews, networking, and guest talks from industry professionals.

ABET LEARNING OUTCOMES

- Ability to identify, formulate, and solve complex engineering problems.
- Application of engineering design principles considering public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- Effective communication skills for diverse audiences.
- Recognition of ethical and professional responsibilities, considering the impact of engineering solutions.
- Ability to work effectively in teams, providing leadership and collaboration.
- Capability to conduct experiments, analyze data, and use engineering judgment to draw conclusions.
- Ability to acquire and apply new knowledge using appropriate learning strategies.

REQUIRED TEXT

- Class Notes / Instructor-assigned reading materials
- IEEE Code of Ethics
- IISE Code of Ethics

Recommended Reading:

- *The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer*, Jeffrey Liker
- *Scrum: The Art of Doing Twice the Work in Half the Time*, Jeff Sutherland

ASSESSMENT

- Attendance (Lecture/Charter Check in): 20%
    - Weekly Tag - Charter: 10%
    - 5 topics case study: 10%
- Quiz: 10%
- Mid-term Exam: 20%
- Final Project Delivery: 50%
    - Oral Presentation: 25%
    - Final Proposal: 25%

Grading Scale: A: 93—100; A-: 90—92; B+: 87—89; B: 83—86; B-: 80—82; C+: 77—79; C: 73—76; C-: 70—72; D: 60—69; F: below 60

Attendance & Check-ins: There are 5 case studies to record attendance, including engineering ethics, design standards, trade-off analysis, ergonomics and risk assessment. Essential for participation, project guidance, and professional development. Absences require a 24-hour notice.

Quiz and Mid-term Exam: Assess Engineering Ethics, IEEE Project Design Standards, Project Management, Lean Management, etc, covering all theories.

Final Oral Presentation for Proposal: Presented to the department, with interactive discussions and peer-reviewed references.
Final Report for Proposal: A structured technical report detailing project development and outcomes, integrating real-world industry applications.

Late Submission Policy:

- 10% off after 24 hours
- 20% off after 48 hours
- 50% off after 72 hours
- No credit beyond 72 hours

ENGR 470 - Capstone Project I (Fall 2026)

Week	Date	Day	Session Type	Topic / Activity	Notes / Due
W1	8/18	Tue	Lecture	L1-Course Overview & Capstone I Objectives - Engineering Design	Review of ABET outcomes, deliverables, and weekly-tag / lecture format for the semester
W1	8/20	Thu	Weekly Tag	P1-Introductions of projects	Client Pitch Day
W2	8/25	Tue	Lecture	L2.1-Engineering Ethics & Case Study	
W2	8/27	Thu	Weekly Tag	P2-Capstone topic brainstorming and discussion, Q&A	Submit your Top 3 choices on OAKS
W3	9/1	Tue	Lecture	L2.2-Engineering Ethics Case Study	(Review, case studies, practice, previous & current)
W3	9/3	Thu	Weekly Tag	P3-Capstone project assignment and announce	Class & Email Announce to Capstone Committee
W4	9/8	Tue	Lecture	L3.1-IEEE Project Design Standards **ABET Quiz**	
W4	9/10	Thu	Weekly Tag	P4-Project Progress based on Charter	Submit the Charter in WeeklyTag on OAKS
W5	9/15	Tue	Lecture	L3.2-IEEE Project Design Standards in capstones (previous & current)	(Review, case studies, practice, previous & current)
W5	9/17	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter
W6	9/22	Tue	Lecture	L4.1-Trade-off Analysis & Design Considerations Lean Management & Continuous Improvement (Kaizen)	
W6	9/24	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter
W7	9/29	Tue	Lecture	L4.2-Trade-off Analysis & Design Considerations (previous & current); Mid-term Review	(Review, case studies, practice, previous & current)
W7	10/1	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter

Week	Date	Day	Session Type	Topic / Activity	Notes / Due
W8	10/6	Tue	Lecture / Exam	Mid-term Exam	
W8	10/8	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter
W9	10/13	Tue	***NO CLASS***	*Fall Break - College Calendar*	
W9	10/15	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter
W10	10/20	Tue	Lecture	L5.1-Ergonomics	New topic: leading and motivating a capstone team
W10	10/22	Thu	Weekly Tag	Project Delivery and report, material list	Plans / next steps
W11	10/27	Tue	Lecture	L5.2-Ergonomics & Case Study;	(Review, case studies, practice, previous & current)
W11	10/29	Thu	Weekly Tag	Project Delivery and report, material list	Check in Charter
W12	11/3	Tue	***NO CLASS***	*Election Day - College Calendar*	
W12	11/5	Thu	**MILESTONE 1 (Weekly Tag)**	**Conceptual prototype concept + hardware/materials list to test out presented to class**	ALL STUDENTS REQUIRED - project process, plans, roadblockers, next steps
W13	11/10	Tue	Lecture	L6.1-Risk Assessment	
W13	11/12	Thu	Weekly Tag	Milestone 1 feedback follow-up	Check in Charter
W14	11/17	Tue	Lecture	L6.2-Risk Assessment & Case Study (previous & current)	(Review, case studies, practice, previous & current)
W14	11/19	Thu	Weekly Tag	Proposal presentation prep	Check in Charter
W15	11/24	Tue	***NO CLASS***	Final Proposal Presentation Prep	
W15	11/26	Thu	***NO CLASS***	*Thanksgiving Holiday - College Calendar*	
W16	12/1	Tue	Lecture	Dryrun: Final Proposal Presentation, Course Wrap-Up & Final Course Evaluations	Last day of classes; final polish/rehearsal ahead of Final Pitch Day (final exam period)
FINALS	12/5	Sat	**FINAL PITCH**	**Final Pitch Day - Final Proposal Presentations**	**3:30 p.m.-5:30 p.m.**, per CofC Fall 2026 Exam Schedule for classes meeting TR at/after 5:30 p.m. but before 6:45 p.m. (class meets 6:30-7:45 p.m.);

This syllabus provides a comprehensive guide to the course expectations, topics, and assessment structure. Students should stay engaged, participate actively, and take advantage of the professional development opportunities provided throughout the semester.

Key No-Class Dates

ENGR 470 - Fall 2026 (College of Charleston calendar):

- Monday-Tuesday, Oct 12-13 - Fall Break;
- Tuesday, Nov 3 - Election Day;

- Wednesday-Sunday, Nov 25-29 - Thanksgiving Holiday (college closed Nov 26-29);
- Tuesday, Dec 1 - last day of full-semester classes;
- Wednesday, Dec 2 - Reading Day;
- Saturday, Dec 5 - Final Pitch Day

ENGR 471 - Capstone Project II (Spring 2027)

Week	Date	Day	Session Type	Topic / Activity	Notes / Due
W1	1/12	Tue	Lecture	Course Overview & Capstone II Objectives L7.1-Lean Management	Transition from proposal (Capstone I) to execution; EXPO overview
W1	1/14	Thu	Weekly Tag	Team status check-in	Review of Capstone I proposal & prototype hand-off; sponsor/mentor confirmation; attendance required for all students
W2	1/19	Tue	Lecture	L7.2-Lean Management	(Review, case studies, practice, previous & current)
W2	1/21	Thu	Weekly Tag	Detailed design progress	Materials/resources needed
W3	1/26	Tue	Lecture	L8.1-IP & Commercialization Verification & Validation Planning; Test Plan Development	
W3	1/28	Thu	Weekly Tag	Materials/parts procurement status	Roadblockers; next plans
W4	2/2	Tue	Lecture	L8.2-IP & Commercialization (previous & current) capstone	(Review, case studies, practice, previous & current)
W4	2/4	Thu	Weekly Tag	Prototype build progress	Roadblockers; next plans
W5	2/9	Tue	Lecture	L9.1-Manufacturing and Scale	
W5	2/11	Thu	Weekly Tag	Integration progress	Materials requests
W6	2/16	Tue	Lecture	L9.2-Manufacturing and Scale (previous & current) capstone	(Review, case studies, practice, previous & current)
W6	2/18	Thu	**MILESTONE II (Weekly Tag)**	**Working prototype demonstrated with initial test/validation results**	ALL STUDENTS REQUIRED - hardware/software integration status, roadblockers, next plans
W7	2/23	Tue	Lecture	L10.1-Customer Development & Lean Business Canvas	
W7	2/25	Thu	Weekly Tag	Mentor / stakeholder communication updates	Milestone II feedback follow-up
W8	3/2	Tue	***NO CLASS***	*Spring Break - College Calendar*	
W8	3/4	Thu	***NO CLASS***	*Spring Break - College Calendar*	
W9	3/9	Tue	Lecture	L10.2-Customer Development & Lean Business Canvas (previous & current) capstone Leadership in Engineering Teams	(Review, case studies, practice, previous & current)

Week	Date	Day	Session Type	Topic / Activity	Notes / Due
W9	3/11	Thu	Weekly Tag	Build progress post-break	Materials requests; roadblockers
W10	3/16	Tue	Lecture	L11.1-Entrepreneur & Lean Startup	New topic: go-to-market thinking, business model, commercialization
W10	3/18	Thu	Weekly Tag	Testing / validation progress	
W11	3/23	Tue	Lecture	L11.1-Entrepreneur & Lean Startup (previous & current) capstone Entrepreneurship & Commercialization Fundamentals	(Review, case studies, practice, previous & current)
W11	3/25	Thu	Weekly Tag	Customer/stakeholder feedback integration	Ties in with Customer Development lecture
W12	3/30	Tue	Lecture	L12-Job Readiness-Invited Alumni	AJ Smith from Tesla
W12	4/1	Thu	**MILESTONE III (Weekly Tag)**	**Draft final poster, report and final presentation, reviewed for feedback**	ALL STUDENTS REQUIRED - project process, plans, roadblockers, next steps; final polish ahead of EXPO and Final Pitch Day
W13	4/6	Tue	Lecture	L13.2-Technical Report (previous & current)	Incorporating Milestone III feedback
W13	4/8	Thu	Weekly Tag	EXPO / report progress	Roadblockers
W14	4/13	Tue	Lecture	EXPO Poster / Report Preparation	
W14	4/15	Thu	Weekly Tag	DryRun: Final presentation prep	Q&A practice
W15	4/20	Tue	Lecture	Final Oral Presentation Prep (structure, Q&A technique)	Last Tuesday lecture of the semester
W15	4/22	Thu	Weekly Tag	DryRun:Course Wrap-Up & Final Course Evaluations	Last Tue/Thu class meeting; final report/EXPO deliverables due ahead of Final Pitch Day (final exam period)
FINALS	5/1 (projected)	Sat	**FINAL EXAM PERIOD**	**Final Pitch Day - Final Technical Presentations**	Spring 2027 exam-block schedule not yet posted by the Registrar (as of Jul 2026);